Day 014 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Model a Rational-Number Situation

Standards: 6.NS.B.2, 6.NS.B.3

Learning target: I can model a Rational-Number Situation, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder. Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 9, 12

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01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder. Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use long division: 648 ÷ 9. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 64 ÷ 9: write 7, subtract 63, remainder 1

3. bring down the next digit: 18 ÷ 9: write 2, subtract 18, remainder 0.

4. Check: 9 × 72 = 648.

Answer: 72

Worked example 2

Use the standard algorithm: 2.56 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 32/5
B. 64
C. 27/5
D. 37/5

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 10: 25.6 ÷ 4 = 6.4.

3. Check: 0.4 × 6.4 = 2.56.

Answer: A - 32/5

Day 014 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 129 ÷ 3. Show the algorithm and check.

02 / Constructed response

Use the standard algorithm: 47.4 + 6.7. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 65.4 × 3.5. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 9.36 ÷ 1.2. Show your regrouping or division steps.

05 / Constructed response

Use long division: 8,178 ÷ 29. Show the algorithm and check.

06 / Constructed response

Use the standard algorithm: 4.02 + 4. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 5.57 × 0.15. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 0.825 ÷ 0.15. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.

A. 13/2
B. 11/2
C. 9/2
D. 55

09 / Model and constructed response

Use long division: 8,443 ÷ 23. Show the algorithm and check.

10 / Constructed response

A runner walks 1.31 km in the morning and 0.27 km later. What is the total distance?

Day 014 | Unit 01 | Answer key

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1. 43

12 ÷ 3: write 4, subtract 12, remainder 0; bring down the next digit: 9 ÷ 3: write 3, subtract 9, remainder 0. Check: 3 × 43 = 129.

6.NS.B.2 | Constructed response

2. 54.1

Align decimal points; 47.4 + 6.7 = 54.1.

6.NS.B.3 | Constructed response

3. 228.9

Compute the whole-number product and restore decimal places: 65.4 × 3.5 = 228.9.

6.NS.B.3 | Constructed response

4. 7.8

Multiply both numbers by 10: 93.6 ÷ 12 = 7.8. Check: 1.2 × 7.8 = 9.36.

6.NS.B.3 | Constructed response

5. 282

81 ÷ 29: write 2, subtract 58, remainder 23; bring down the next digit: 237 ÷ 29: write 8, subtract 232, remainder 5; bring down the next digit: 58 ÷ 29: write 2, subtract 58, remainder 0. Check: 29 × 282 = 8178.

6.NS.B.2 | Constructed response

6. 8.02

Align decimal points; 4.02 + 4 = 8.02.

6.NS.B.3 | Constructed response

7. 0.8355

Compute the whole-number product and restore decimal places: 5.57 × 0.15 = 0.8355.

6.NS.B.3 | Constructed response

8. B - 11/2

Multiply both numbers by 100: 82.5 ÷ 15 = 5.5. Check: 0.15 × 5.5 = 0.825.

6.NS.B.3 | Multiple choice

9. 367 R 2

84 ÷ 23: write 3, subtract 69, remainder 15; bring down the next digit: 154 ÷ 23: write 6, subtract 138, remainder 16; bring down the next digit: 163 ÷ 23: write 7, subtract 161, remainder 2. Check: 23 × 367 + 2 = 8443.

6.NS.B.2 | Model and constructed response

10. 1.58 km

Align decimal points; 1.31 + 0.27 = 1.58.

6.NS.B.3 | Constructed response